COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	October 21, 2026	**DATE SUBMITTED:**	October 21, 2026
NAME:	Clifford Lee C. Doplayna Joel Almirol	**INSTRUCTOR:**	Engr. Jimlord Quejado

Hands-on Activity 14.1
Algorithm Complexity

1. PROCEDURE OUTPUT

	Best Case	Worst Case	Analysis
Algorithm 1 (Bubble Sort)	**O(n^2)**	**O(n^2)**	The program uses two fixed nested loop structures without an early exit flag. Consequently, whether the array is sorted or reversed, it performs every comparison step. The running time scales quadratically with respect to input size n.
Algorithm 2 (Merge Sort)	**O(n log2 n)**	**O(n log2 n)**	The program uses a recursive divide-and-conquer strategy. It repeatedly splits the array into two equal parts until length reaches 1, then merges them back in sorted order. Splitting takes logarithmic steps while merging takes linear time, resulting in consistent linearithmic complexity.

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

	Algorithm 1	Algorithm 2

Best Case		
Worst Case	N = 100 | Bubble Sort Worst: 0.0477 ms N = 1000 | Bubble Sort Worst: 5.2045 ms N = 10000 | Bubble Sort Worst: 470.747 ms	Merge Sort Worst: 0.0505 ms Merge Sort Worst: 0.498 ms Merge Sort Worst: 5.089 ms

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

Analysis:

The results from the empirical analysis in Table clearly indicate the distinctive performances of quadratic (O(n^2) and linearithmic (O(n \log_2 n)) sorting algorithms, depending on how large the input gets. For example, at a relatively small size of input N=100, Bubble Sort 0.0477 text is somewhat faster than Merge Sort 0.0505 text , owing to its simplicity and negligible function call overhead. But when the size of input becomes N=1,000, the execution time of Merge Sort becomes around ten times faster than that of Bubble Sort 0.498 text vs. 5.2845 text. At an even higher size of N=10,000, the execution time of Bubble Sort sharply rises to 470.747 text due to its inherent structure of fixed nested loop comparisons, but that of Merge Sort remains 5.089 text only—that is 92 times faster than Bubble Sort.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO A: Contrast and compare the two searching techniques used in class. The required deliverable for this activity would be the theoretical and empirical analysis.

Algorithm	Pseudocode	Best Time	Worst	Space Complexity
Linear Search	FUNCTION linearSearch(list, target): FOR EACH item IN list DO IF item == target THEN RETURN index END FOR RETURN -1	**O(1)**	**O(n)**	**O(1)**

	END FUNCTION			
Iterative Binary Search	FUNCTION binarySearch(arr, n, target): SET low = 0, high = n - 1 WHILE low <= high DO SET mid = (low + high) / 2 IF arr[mid] == target THEN RETURN mid ELSE IF target < arr[mid] THEN high = mid - 1 ELSE low = mid + 1 END WHILE RETURN -1 END FUNCTION	**O(1)**	**O(log n)**	**O(1)**
Recursive Binary Search	UNCTION recursiveBinarySearch(arr, low, high, target):	**O(1)**	**O(log n)**	**O(log n)**

	IF low > high THEN RETURN -1 SET mid = low + (high - low) / 2 IF arr[mid] == target THEN RETURN mid ELSE IF target < arr[mid] THEN RETURN CALL recursiveBinarySearch(arr, low, mid - 1, target) ELSE RETURN CALL recursiveBinarySearch(arr, mid + 1, high, target) END FUNCTION			
Binary Search	UNCTION binarySearchLL(head, target): SET start = head, last = NULL WHILE start != last DO	**O(1)**	**O(n)**	**O(1)**

	SET mid = CALL getMiddle(start, last) IF mid == NULL THEN RETURN NULL IF mid.data == target THEN RETURN mid ELSE IF mid.data < target THEN start = mid.next ELSE last = mid END WHILE RETURN NULL END FUNCTION			

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000	N = 1000 | Binary Search: 0.0002 ms	Linear Search: 0.0032 ms
10,000	N = 10000 | Binary Search: 0.0002 ms	Linear Search: 0.0307 ms
100,000	N = 100000 | Binary Search: 0.0003 ms	Linear Search: 0.3067 ms

Analysis:

As seen from the empirical results given below, it becomes clear that there is a significant difference between the efficiency of Linear Search and Binary Search in the worst case situation, and this actually proves the theoretical complexity of these two algorithms. When increasing the value of the input, N, from 1,000 to 10,00 and 100,000 by 10 times, the time required to execute Linear Search algorithm increases linearly (in O(N)) to 0.0032\text{ ms}, 0.0307\text{ ms} and 0.3067\text{ ms} respectively due to sequential check of all elements whether the desired one is in the last position or does not exist. At the same time, the time required for Binary Search is almost constant and increases insignificantly to 0.0002\ text - 0.0003\text due to decreasing of the search range by half. It is worth mentioning that Binary Search needs not more than \lceil \log_2(N) \rceil comparisons (around 17 comparisons for N = 100,000).

Conclusion:

Doplayna:

After finishing this activityI learned how to compute time and space complexities using asymptotic Big-O notation, analyzing how input size n and dominating operational terms directly dictate algorithmic execution speed regardless of underlying hardware platforms. The things that I need to improve is my empirical testing skills by setting up precise timing benchmarks and deepen my understanding of analyzing complex non-polynomial algorithms.

Almirol:

I have been able to understand the difference between searching and sorting algorithms both practically and theoretically when using different input sizes. Though some algorithms such as linear search and bubble sort are simple to execute, their inefficiency due to O(n) and O(n^2) time complexity makes them unsuitable for big inputs. On the other hand, some divide and conquer algorithms like binary search of O(\log n) and merge sort of O(n \log n) remain efficient even when input size increases. It is clear from the above that picking the right algorithm will depend much on the type and size of the data since Binary search is faster with sorted input while linear search is fast with small or unsorted input.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

COMPUTER ENGINEERING

QUEZON CITY

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

Rubrics:

Rubric for SO 7 (6)

Criteria	Ratings						Pts
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24